

Amrita School of Engineering, Bengaluru-35

23MAT206 – Optimization Techniques

Lab Practice Sheet-1

Identifying definiteness of matrices using eigenvalues, and use of Hessian matrix to identify concavity of the surfaces using MATLAB

Surface Plots

The parametric representation of any surface is given using two parameters. If a surface is given by the equation, $Z=f(X,Y)$, the parametric representation of this surface can be taken as $r(u,v)=(u,v,f(u,v))$. For example, parametric representation of the plane $z=x+y$ is $r(u,v)=[u,v,u+v]$.

```
% surface plot of the paraboloid z=x^2+y^2 using command 'surf'
[X,Y] = meshgrid(-10:0.5:10,-10:0.5:10); % returns 2-D grid coordinates between -10<x,y<10
surf(X,Y,X.^2+Y.^2)
xlabel('x')
ylabel('y')
zlabel('z')
title('Surface z=x^2+y^2')

% surface plot of the cone z=Root(x^2+y^2) using command 'mesh'
[X,Y] = meshgrid(-2:0.01:2); % returns 2-D grid coordinates for x and y between -2
and 2
Z=(X.^2+Y.^2).^(0.5);
mesh(X,Y,Z)
xlabel('x')
ylabel('y')
zlabel('z')
title('Cone z=Root(x^2+y^2)')
```

Positive Definite Matrix:

A symmetric matrix S is defined to be a positive definite matrix if for any point x in the search space the quantity (energy) $x^T S x > 0$ for all $x \neq 0$. Geometrically we can understand this if graph of $f(x) = x^T S x$ curves up. If $f(x) = x^T S x$ curves down, the symmetric matrix S is negative definite.

Example: $S = \begin{bmatrix} 3 & 0 \\ 0 & 5 \end{bmatrix}$ is positive definite as $f(x) = x^T S x = 3x^2 + 5y^2$ curves up. It can be verified by plotting the surface $z = 3x^2 + 5y^2$ in MATLAB.

Contour curves/ level curves

```
x = linspace(-2*pi,2*pi); y = linspace(0,4*pi);
[X,Y] = meshgrid(x,y);
Z = sin(X)+cos(Y);
contour(X,Y,Z) % contour curves for the function f(x,y)=sinx+cosy
```

Contour curves/ level curves with value of the function written on the curves

```
x = linspace(-2*pi,2*pi); y = linspace(0,4*pi);
[X,Y] = meshgrid(x,y);
Z = sin(X)+cos(Y);
contour(X, Y, Z, 'ShowText', 'on')
```

contour(x,y,z,50) can be used to get fixed number of contour curves/ level curves

➤ contour(X,Y,Z,'--') can be used to get contour curves in -- style

Surface plots with contour curves

```
x = linspace(-2*pi,2*pi); y = linspace(0,4*pi); [X,Y] = meshgrid(x,y);
Z = sin(X)+cos(Y); surf(X,Y,Z);
shading interp
```

Gradient using MATLAB

```
syms x y z; gradient(x^2 + 3*y^2-5*z, [x, y, z])
```

Plot of surface, level curves and gradient vectors of a scalar function f at different points on the level curves

```
[x,y] = meshgrid(-2:.2:2,-2:.2:2); z = x.^2 + y.^2;
figure
surf(x,y,z);
figure
[px,py] = gradient(z); contour(x,y,z,'ShowText','on');
hold on
quiver(x,y,px,py);
hold off;
axis image
% Here surface plot will appear in one figure window and the gradient vectors in another
```

Hessian matrix using MATLAB

```
syms x y z; hessian(x*y + 2*z*x, [x, y, z])
```

Practice Questions

1. Plot the surface given by $z = \sin(x + y)$, $-1 \leq x, y \leq 1$.
2. Plot the surface of the cone given by $z = \sqrt{6x^2 + 9y^2 - 5}$, $-5 \leq x, y \leq 5$.
3. Plot the surface $z = \sin(x + y) - x^2 - xy + y^2$, $-2\pi \leq x, y \leq 2\pi$.
4. By plotting the surface $f(x) = x^T S x$ identify if the given matrices are positive definite, negative definite or indefinite. Also verify by finding the eigenvalues of each of these matrices.

(i) $A = \begin{bmatrix} 3 & 2 \\ 2 & 5 \end{bmatrix}$ (ii) $B = \begin{bmatrix} 9 & 4 \\ 4 & 6 \end{bmatrix}$ (iii) $C = \begin{bmatrix} -5 & 3 \\ 3 & -8 \end{bmatrix}$ (iv) $D = \begin{bmatrix} -7 & -1 \\ -1 & -4 \end{bmatrix}$

5. Generate a 6 by 2 matrix A. Find $S = A^T A$ and verify that $S = A^T A$ is a positive definite matrix by:
 - a. by finding the eigenvalues of S.
 - b. by plotting the surface $f(x) = x^T S x$.
6. Find the gradient and Hessian of the following scalar function using the MATLAB command
 - a. $f(x, y, z) = 4x^3 e^{xy} - y^3 z$
 - b. $g(x, y) = e^{xy}(x + y)$ at $P: (-1, 1)$
 - c. $h(x, y, z) = (x^2 + y^2)/(x^2 - y^2)$
7. Consider the function: $f(x, y) = 1/(|x| + |y| + 0.5)$. Plot 15 level curves for the function $f(x, y)$ in the rectangular domain $-5 \leq x, y \leq 5$.
8. Consider the function: $f(x, y) = 4x^2 + 9y^2 - 72$.
 - a) Plot the level curves for $f(x, y)$ with function values written in each curve in $-10 \leq x, y \leq 10$
 - b) Find the gradient and hessian of the function using 'syms'.
 - c) Plot the gradient vectors at some of the points on the level curves.
9. Consider the function $f(x, y) = \sin x + \cos y + 3e^{x^2 - y^2}$.
 - d) Show the surface plot of the function in one figure and the gradient vectors at some points on level curves in another figure.
 - e) Looking at the surface plot and from the figure with function values on level curves, try to find a minimum and a maximum for the function.
10. Consider the function: $f(x, y) = x e^{x^2 - y^2}$.
 - a) Plot the surface given for x and y varying from -2 to 2
 - b) Write the General expression for gradient and Hessian for the function
 - c) Evaluate Hessian at the points given in the following figures
 - d) Find eigenvalues of those hessian matrices and decide the nature of curvature at the points where it is evaluated. (curved upward, downward, neither)

